UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

# Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

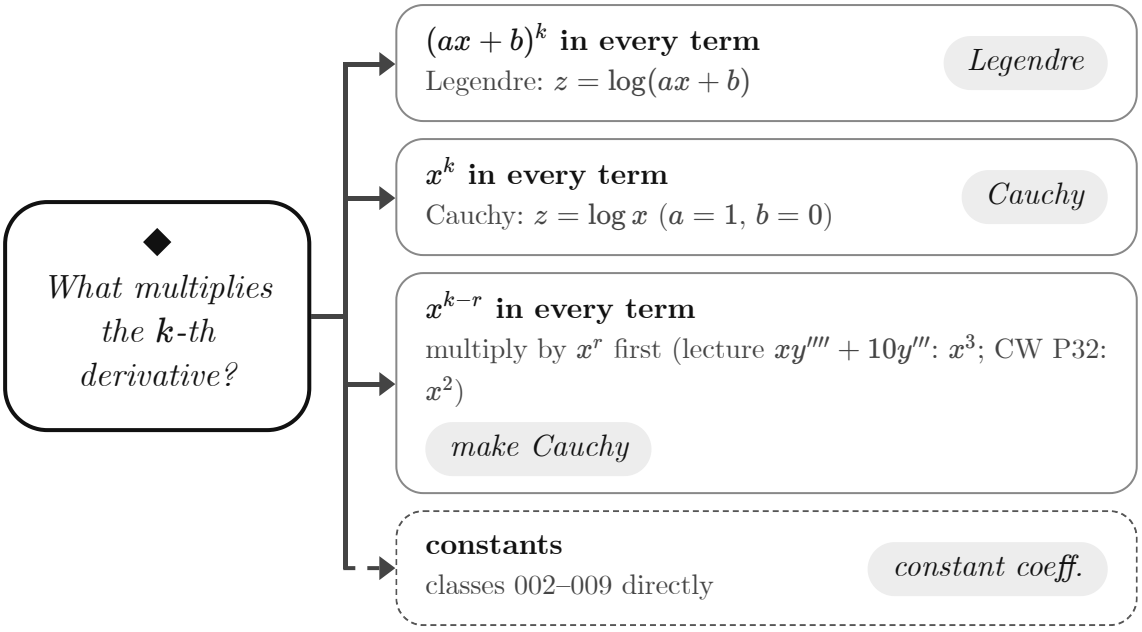
Class 011 (Legendre) · Legendre's and Cauchy's linear differential equations · compiled 17 October 2026

| | |
|---|---|
| ESA ★★★ | Asked repeatedly in past ESA papers |
| ESA ★★ | Asked in past ESA papers or computed in class-work problems |
| ESA ★ | Conceptual: taught in class, not examined so far |
| ADDED FOR CLARITY | Standard context that is not in the class material |
| TRANSCRIPT ONLY | Said in the lecture but not on any slide |
| VIDEO ONLY | Seen only in a frame of the class video |
| NOT ON CURRENT SLIDES | From older material or past papers, not this year's slides |
| S011L·8 | Unit 2 deck 011 “Legendre's differential equation” (SESSION 9), slide 8 (rendered .pptx) |
| T011L 00:05:20 | AV-summary transcript “Legendre differential equation” (21.9 min, listed first), segment start |
| T011Lb 00:12:26 | AV-summary transcript “Legendre differential equations_Continued” (23 min, second), segment start |
| CW U2 P30 | Unit 1 & 2 class-work sheet, Unit 2 problem 30 |
| ESA Jul-23 3a | Past paper: UE22MA141A July 2023, question 3.a (Jan-24 s1...s4 = UE23MA141A Jan 2024 sets 1–4) |

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2. Figure F2.1 · Solving a Legendre (or Cauchy) equation
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4. Figure F3.2 · ESA Jul-23 3a = May-24 3a
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7. Every problem in this pack, reduced to $z$
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# 1 Linear equations with variable coefficients: Legendre's and Cauchy's forms

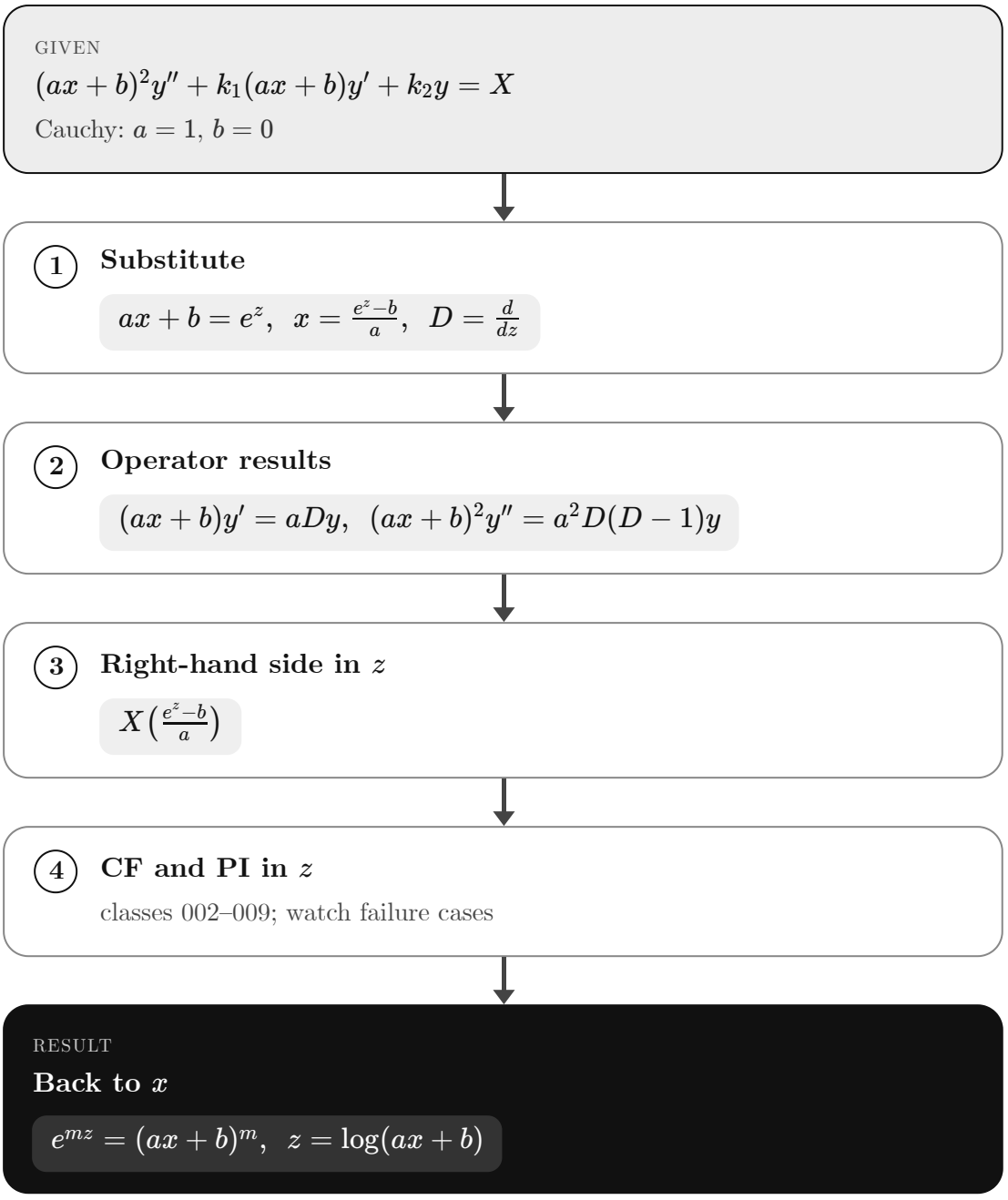



***Remark.*** The test is the same for all: the power of the factor equals the order of the derivative.

**Figure F1.1.** Which form is it? ESA ★★★

S011L·4 · T011L 00:04:06 · T011Lb 00:18:25 · CW U2 P32

# 2 The substitution $ax + b = e^z$ and the operator results




***Remark.*** In steps 2–4 the variable is $z$: a failure case gives $z\,e^{kz}$.

***Exam trap.*** $a^2D(D-1)$, not $a^2D^2$ and not $D(D-1)$.

**Figure F2.1.** Solving a Legendre (or Cauchy) equation ESA ★★★

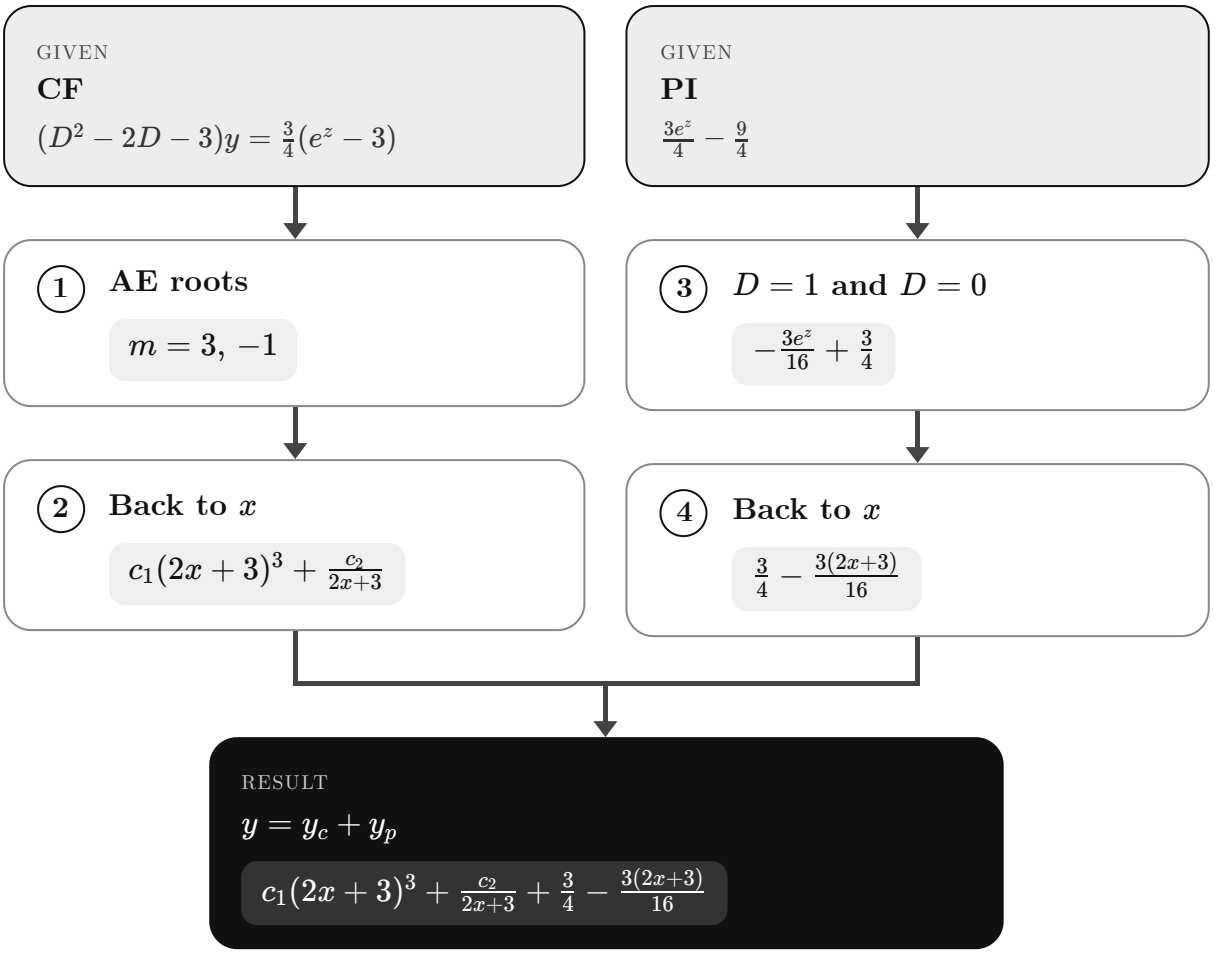
S011L·5 · S011L·6 · S011L·7 · T011L 00:13:26 · T011Lb 00:12:12

## Legendre and Cauchy side by side

| | Legendre | Cauchy |
|---|---|---|
| **Coefficient of $y^{(k)}$** | $k_{n-k}(ax+b)^k$ | $k_{n-k}x^k$ |
| **Substitution** | $ax+b=e^z$, $z=\log(ax+b)$ | $x=e^z$, $z=\log x$ |
| $\frac{dz}{dx}$ | $\frac{a}{ax+b}$ | $\frac{1}{x}$ |
| **First order** | $(ax+b)y'=aDy$ | $xy'=Dy$ |
| **Second order** | $(ax+b)^2y''=a^2D(D-1)y$ | $x^2y''=D(D-1)y$ |
| **Third order** | $a^3D(D-1)(D-2)y$ | $D(D-1)(D-2)y$ |
| $e^{mz}$ **in** $x$ | $(ax+b)^m$ | $x^m$ |
| **On the slides?** | yes (011L·5–10) | form only (011L·4); rest transcript only |

S011L·4 · S011L·5 · S011L·6 · T011L 00:18:30 · T011L 00:19:07

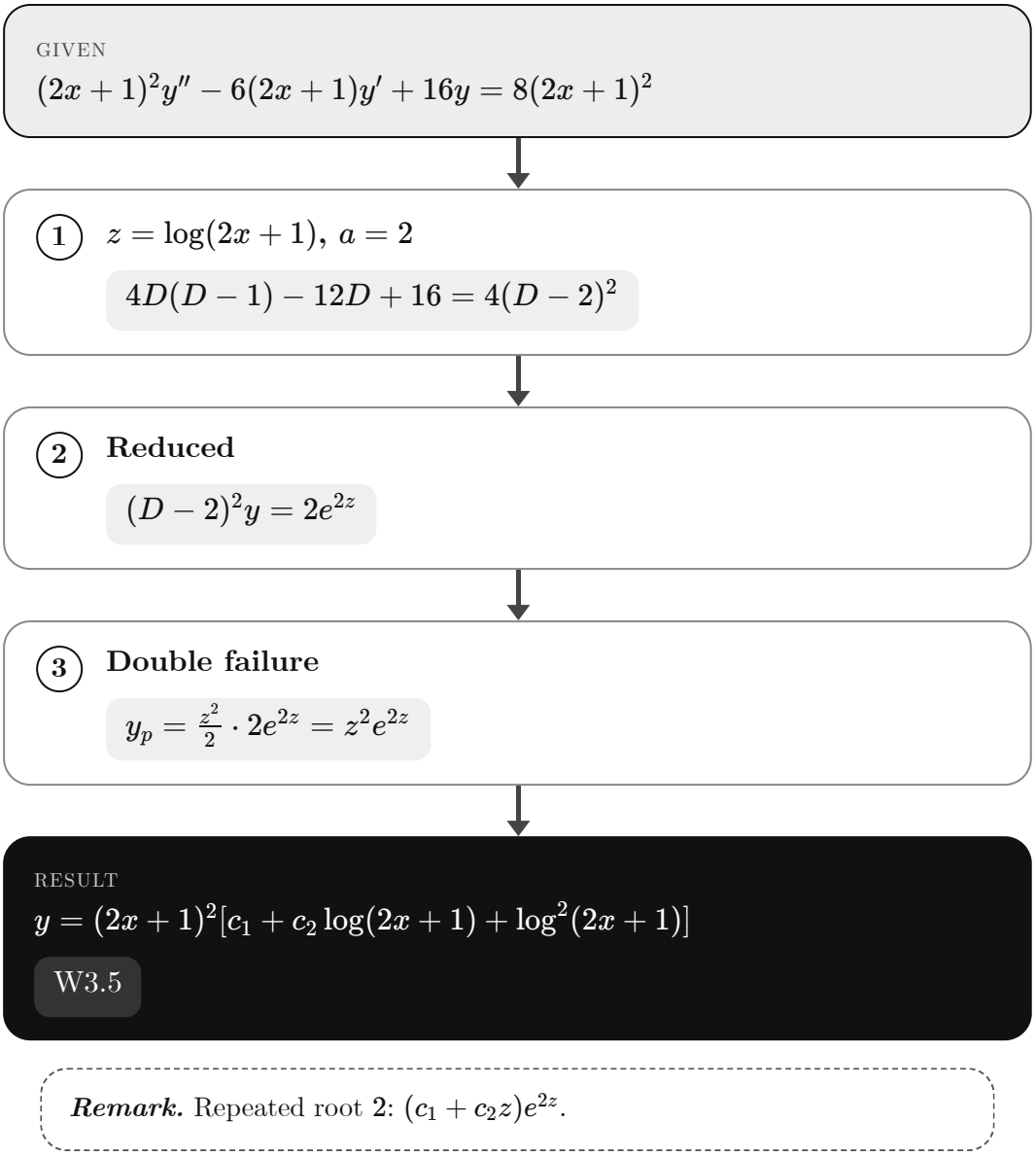
# 3 Legendre problems: the slide example, CW P29–P30 and five ESA questions




***Remark.*** Worked in full as W3.1.

**Figure F3.1.** Slides 011L·8–10: $(2x+3)^2y''-2(2x+3)y'-12y=6x$
ESA ★★★

S011L·8 · S011L·9 · S011L·10




***Remark.*** Repeated root 2: $(c_1 + c_2 z)e^{2z}$.

**Figure F3.2.** ESA Jul-23 3a = May-24 3a ESA ★★★

ESA Jul-23 3a · ESA May-24 3a

### Every problem in this pack, reduced to $z$

| Problem | Equation in $z$ | Roots | PI in $x$ |
|---|---|---|---|
| **Slide (W3.1)** | $(D^2 - 2D - 3)y = \frac{3}{4}(e^z - 3)$ | $3, -1$ | $\frac{3}{4} - \frac{3(2x+3)}{16}$ |
| **CW P29 (W3.2)** | $(D^2 + 1)y = 2\sin z$ | $\pm i$ | $-\log(1+x)\cos\log(1+x)$ |
| **Dec-23 (W3.3)** | $(D^2 + 1)y = \sin 2z$ | $\pm i$ | $-\frac{1}{3}\sin(2\log(1+x))$ |
| **CW P30 (W3.4)** | $(D^2 - 4)y = \frac{e^{2z}-1}{27}$ | $\pm 2$ | $\frac{(3x+2)^2\log(3x+2)+1}{108}$ |
| **Jul-23 (W3.5)** | $(D-2)^2y = 2e^{2z}$ | $2, 2$ | $(2x+1)^2\log^2(2x+1)$ |
| **Jan s1 (W3.6)** | $4(D^2 - 2D - 3)y = 3e^z + 2$ | $3, -1$ | $-\frac{3(2x+1)}{16} - \frac{1}{6}$ |

| Problem | Equation in $z$ | Roots | PI in $x$ |
|---|---|---|---|
| **Jan s2 (W3.7)** | $(D-1)^2y = 3e^z - 2$ | $1, 1$ | $\frac{3}{2}(x+2)\log^2(x+2) - 2$ |
| **Jun-24 (W3.8)** | $(4D^2-2D-2)y = 2e^{2z} + 3e^z + 4$ | $1, -\frac{1}{2}$ | $\frac{(2x-1)^2}{5} + \frac{(2x-1)\log(2x-1)}{2} - 2$ |
| **Lecture (W4.1)** | $(D^3 - D^2 + 2)y = 10(e^z + e^{-z})$ | $-1, 1 \pm i$ | $5x + \frac{2\log x}{x}$ |
| **CW P31 (W4.3)** | $(D^2 - 2D - 3)y = ze^{2z}$ | $3, -1$ | $-\frac{x^2}{3}(\log x + \frac{2}{3})$ |
| **CW P32 (W4.4)** | $D^2y = 12z$ | $0, 0$ | $2\log^3 x$ |
| **CW P33 (W4.5)** | $(D^2 + 3D + 2)y = e^z + e^{-z}$ | $-1, -2$ | $\frac{x}{6} + \frac{\log x}{x}$ |
| **Jan s3 (W4.6)** | $(D-2)(D-3)y = e^{4z}\sin e^z$ | $2, 3$ | $-x^2 \sin x$ |

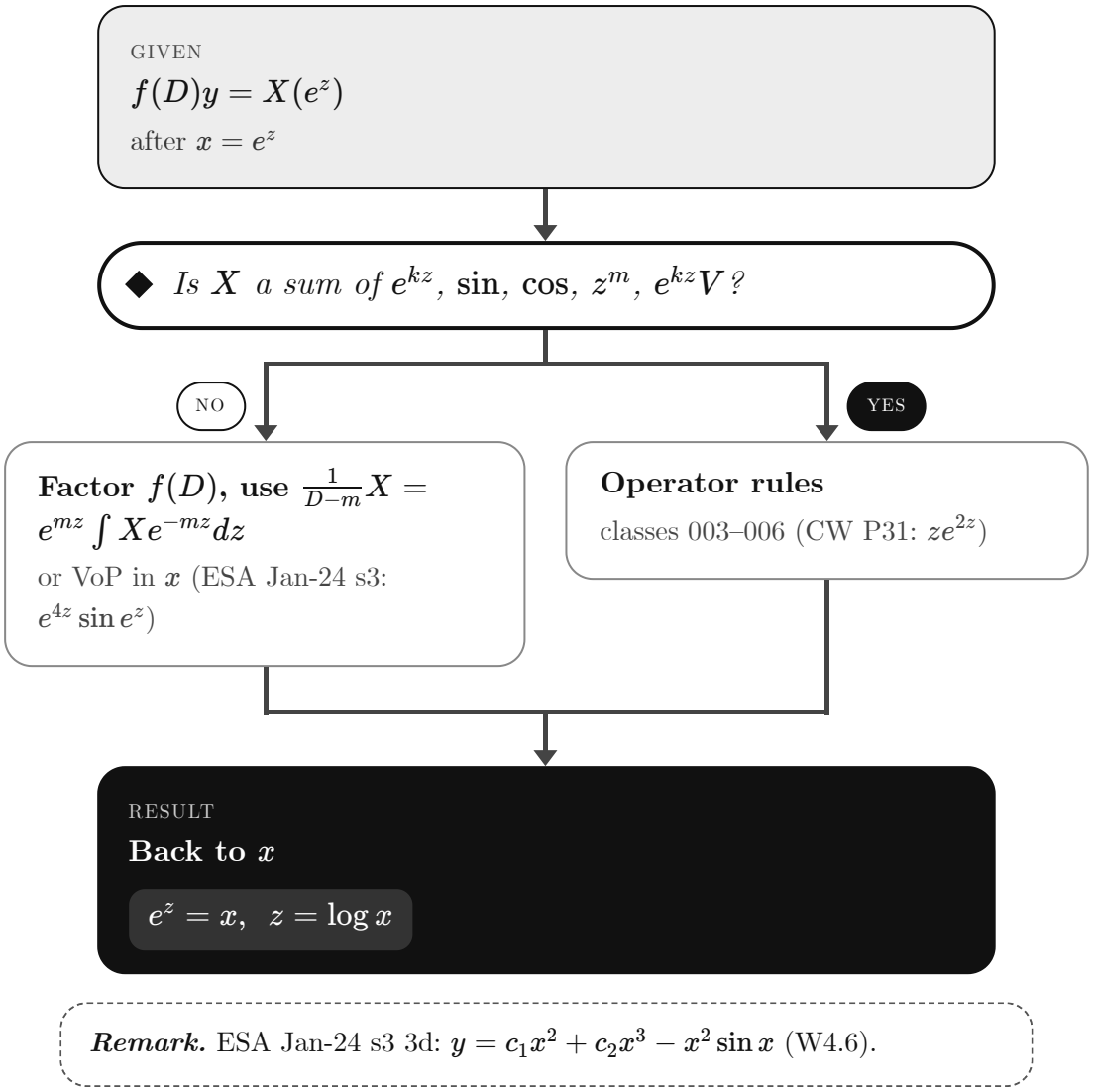
S011L·8 · CW U2 P29 · CW U2 P30 · CW U2 P31 · CW U2 P32 · CW U2 P33 · ESA Dec-23 3b · ESA Jul-23 3a · ESA Jan-24 s1 3c · ESA Jan-24 s2 3c · ESA Jun-24 3b · ESA Jan-24 s3 3d · T011Lb 00:04:59

## Printed answers that are wrong

| Source | Printed | Correct |
|---|---|---|
| **Slides 011L·4–5** | $k_2(\cdot)^{n-2}\frac{d^{n-1}y}{dx^{n-1}}$ | $k_2(\cdot)^{n-2}\frac{d^{n-2}y}{dx^{n-2}}$ |
| **CW P29** | $-\log(1+x) - \cos\log(1+x)$ | $-\log(1+x)\cos\log(1+x)$ |
| **CW P30** | $C_2[(2+3x)^2]$ | $C_2(2+3x)^{-2}$ |
| **CW P31** | $-\frac{x^3}{3}(\log x + \frac{2}{3})$ | $-\frac{x^2}{3}(\log x + \frac{2}{3})$ |
| **CW P32** | $-2(\log x)^2$ | $+2(\log x)^3$ |

CW U2 P29 · CW U2 P30 · CW U2 P31 · CW U2 P32 · S011L·4 · S011L·5

# 4 Cauchy problems: lecture examples, CW P31–P33 and ESA Jan-24 s3 3d




***Remark.*** ESA Jan-24 s3 3d: $y = c_1x^2 + c_2x^3 - x^2\sin x$ (W4.6).

**Figure F4.1.** Is the right-hand side in $z$ a standard form? ESA ★★★

ADDED FOR CLARITY ESA Jan-24 s3 3d · CW U2 P31 · CW U2 P33